

MICHAELA KLIMOVA

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RESEARCH POSITIONS

2025 - present	Postdoctoral Research Associate, Duke University (July 2026 - present) and Northeastern University (2025 - 2026) Subjectivity lab, advisor: Dr. Jorge Morales
2023 - 2025	Postdoctoral Research Associate, Northeastern University Kwon lab for low vision research, advisor: Dr. MiYoung Kwon
2016 - 2017	Predocotrual Research Assistant, University of Sussex Sackler Centre for Consciousness Science, advisor: Dr. Anil Seth

EDUCATION

2017 - 2023	PhD in Psychology: Brain, Behavior, and Cognition, Boston University Advisor: Dr. Sam Ling
2015 - 2016	MSc, Human Cognitive Neuropsychology, University of Edinburgh Thesis advisor: Dr. Elena Gherri
2011 - 2015	BSc (with Honors), Psychology, University of Edinburgh Thesis advisor: Dr. David Carmel

PUBLICATIONS

- Cicero, N. G., **Klimova, M.**, Vinke, L., Ling, S., & Lewis, L. D. (2026). Cortex-specific inversion of visual responses during sleep. *Nature Communications* 17, 7403.
- Wiecek, E., Ramirez, L. D., **Klimova, M.**, & Ling, S. (2026). Spatial frequency tuning follows scale invariance in the human visual cortex. *Journal of Neuroscience*, 46,(4).
- Klimova, M.**, & Kwon, M. (2025). Impact of rod-dominant mesopic conditions on spatial summation and surround suppression in early visual cortex. *Journal of Neuroscience*, 45(21).
- Klimova, M.**, Bloem, I. M., & Ling, S. (2025). How does orientation-tuned normalization spread across the visual field?. *Journal of Neurophysiology* 133(2), 539-546.
- Pan, J., Sun, X., Park, E., Kaufmann, M., **Klimova, M.**, McGuire, J. T., & Ling, S. (2024). The effects of emotional arousal on pupil size depend on luminance. *Scientific Reports*, 14(1), 21895.
- Cicero, N. G., **Klimova, M.**, Lewis, L. D., & Ling, S. (2024). Differential cortical and subcortical visual processing with eyes shut. *Journal of Neurophysiology*, 132(1), 54-60.
- Klimova, M.**, Bloem, I. M., & Ling, S. (2023). Attention preserves the selectivity of feature-tuned normalization. *Journal of Neurophysiology*, 130(4), 990-998.
- Ling, S., & **Klimova, M.** (2022). Davida reorients intermediate visual processing. *Cognitive Neuropsychology*, 39(1-2), 88-91.
- Pan, J., **Klimova, M.**, McGuire, J. T., & Ling, S. (2022). Arousal-based pupil modulation is dictated by luminance. *Scientific Reports*, 12(1), 1390.
- Klimova, M.**, Bloem, I. M., & Ling, S. (2021). The specificity of orientation-tuned normalization within human early visual cortex. *Journal of Neurophysiology*, 126(5), 1536-1546.

Klimova, M., Nishida, S. Y., & Roseboom, W. (2017). Grouping by feature of cross-modal flankers in temporal ventriloquism. *Scientific Reports*, 7(1), 7615.

Manuscripts in preparation:

Penaloza, B., **Klimova, M.**, & Kwon, M. (*in prep*). How dim light, crowding, and aging jointly constrain functional field of view.

CONFERENCE PRESENTATIONS

Klimova, M. & Morales, J. (2026). The role of visual sensitivity and task demands in gaze behavior towards visually uncomfortable stimuli. Poster presented at *The Annual Meeting for Vision Sciences Society*.

Ekhlasi, A., Olsson, E., **Klimova, M.**, Huang, X., Shen, A., Dijkstra, N., Morales, J.*, Peters, M. A. K.* (2026). How does your brain know what's real? Domain adaptation and fMRI approaches to revealing the neural mechanisms of reality monitoring. Poster presented at *The Annual Meeting for Vision Sciences Society*.

Huang, X., Shen, A., Olsson, E., **Klimova, M.**, Ekhlasi, A., Yurdakul, O. C., Peters, M. A. K.*, Morales, J.* (2026). What we talk about when we talk about vividness. Poster presented at *The Annual Meeting for Vision Sciences Society*.

Yurdakul, O. C., Huang, X., Shen, A., Olsson, E., Ekhlasi, A., **Klimova, M.**, Morales, J.*, Peters, M. A. K.* (2026). A novel approach to characterizing the visual dimensions of mental imagery vividness using Gaussian process regression. Poster presented at *The Annual Meeting for Vision Sciences Society*.

Ekhlasi, A., Olsson, E., **Klimova, M.**, Huang, X., Shen, A., Garcia, K., Dijkstra, N., Morales, J.*, Peters, M. A. K.* (2025). Reproducible and open-source fMRI decoding pipelines for arbitrating theories of consciousness. Poster presented at *Society for Neuroscience*.

Klimova, M. & Kwon, M. (2025). How does the early visual cortex cope with mesopic conditions? Evidence for compensatory mechanisms in cortical spatial summation and surround suppression. Poster presented at *The Annual Meeting for Vision Sciences Society*.

Cicero, N. G., **Klimova, M.**, Ling, S., & Lewis, L. (2025). Sleep profoundly alters corticothalamic responses to visual stimuli. Poster presented at *The Annual Meeting for Vision Sciences Society*.

Klimova, M., Wiecek, E., Ramirez, L., & Ling, S. (2024). Spatial frequency tuning in early visual cortex is not scale invariant. Poster presented at *The Annual Meeting for Vision Sciences Society*.

Wiecek, E., Ramirez, L. D., **Klimova, M.**, & Ling, S. (2024). Are visual deficits in amblyopia driven by spatial frequency tuning? Poster presented at *The Annual Meeting for Vision Sciences Society*.

Wiecek, E. K., **Klimova, M.**, Ramirez, L. D., & Ling, S. (2024). Is the loss of spatial resolution in amblyopia linked to a deviation in scale invariance? Poster presented at *The Association for Research in Vision and Ophthalmology*.

Klimova, M., Bloem, I. M., & Ling, S. (2023). Orientation-tuned normalization modulates the gain of visuocortical contrast responses in humans. Poster presented at *The Annual Meeting for Vision Sciences Society*.

Wiecek, E., Ramirez, L. D., **Klimova, M.**, & Ling, S. (2023). Characterizing the relationship between population spatial frequency tuning and receptive field size. Poster presented at *The Annual Meeting for Vision Sciences Society*.

Cicero, N., **Klimova, M.**, Lewis, L., & Ling, S. (2023). Eye closure elicits qualitatively distinct responses within the lateral geniculate nucleus and visual cortex. Poster presented at *The Annual Meeting for Vision Sciences Society*.

Pan, J., **Klimova, M.**, McGuire, J., & Ling, S. (2023). Investigating the interaction between affective arousal and luminance in modulating pupil size. Poster presented at *The Annual Meeting for Vision Sciences Society*.

Wiecek, E. K., **Klimova, M.**, Ramirez, L. D., & Ling, S. (2023). Spatial frequency tuning in early visual cortex in individuals with amblyopia. Paper presented at *The Association for Research in Vision and Ophthalmology*.

Klimova, M., Bloem, I. M., & Ling, S. (2021). Tuned normalization bandwidth is unaltered by attention. Poster presented at *The Annual Meeting for Vision Sciences Society*.

Pan, J., **Klimova, M.**, McGuire, J., & Ling, S. (2021). Luminance dictates arousal-based pupil modulation. Poster presented at *The Annual Meeting for Vision Sciences Society*.

Klimova, M., Bloem, I. M., & Ling, S. (2019). Estimating the bandwidth of tuned normalization within human visual cortex. Poster presented at *The Annual Meeting for Vision Sciences Society*.

AWARDS AND HONORS

2026	FoVea Travel and Networking Award, Vision Sciences Society
2025	Vision Sciences Society NEI Travel Award
2020	Clara Mayo Memorial Fellowship, Boston University
2018	Dean's Fellowship, Boston University
2015 - 2016	The Edinburgh University UK/EU Master's Scholarship

INVITED TALKS AND LECTURES

2024	Invited lecture (graduate-level fMRI methods), Northeastern University
2023	Invited talk, Sinha Lab for Developmental Research, MIT
2021	Invited lecture (BS-level fMRI methods), New Bulgarian University

TEACHING

Fall 2026	Instructor, Laboratory in Sensation and Perception, Northeastern University
2018 - 2022	Teaching Fellow, Experimental Psychology: Perception, Boston University
2017 - 2021	Teaching Fellow, Sensation and Perception, Boston University

MENTORING

2025 - present	Julia McClellan (graduate student, Boston University) Pediatric neuroimaging collaboration, mentoring in fMRI methods
2024 - 2025	Chloe Chung (undergraduate research intern, Northeastern University) Training in eye-tracking and vision testing for low-vision research
2023	DaHae Choi (undergraduate research intern, Northeastern University) Work presented as a poster (RISE Expo, Northeastern University, 2024)

PROFESSIONAL ACTIVITIES, CERTIFICATIONS, AND PUBLIC OUTREACH

Ad-hoc reviewer:	Cerebral Cortex Journal of Computational Neuroscience Brain Topography Scientific Reports iScience
2026	LEADERS Leadership Certificate Program, Northeastern University
2025 - 2026	Panelist, Summer CAMP (Cambridge Access to Mental Health and Psychology) Programme, University of Cambridge

2019 - present	MRI certified advanced operator, Boston University Cognitive Neuroimaging Center
2018 - 2020	Secretary, Computational Neuroscience Student Organization, Boston University
2018	Presenter, Summer Pathways in Science and Engineering (outreach for high school students), Boston University
2016	Presenter, In Touch With Reality (outreach event for the Sackler Centre), Wellcome Collection Museum, London UK
2014 - 2015	Leader, Psychology Peer-Assisted Learning Scheme, University of Edinburgh
2013 - 2014	Helpline volunteer, Nightline Association (confidential student peer support), University of Edinburgh

REFERENCES

Jorge Morales, Ph.D.	Department of Philosophy Department of Psychology and Neuroscience Duke Institute for Brain Sciences Center for Cognitive Neuroscience Duke University <i>jorge.morales@duke.edu</i>
MiYoung Kwon, Ph.D.	Department of Psychology Northeastern University <i>m.kwon@northeastern.edu</i>
Sam Ling, Ph.D.	Department of Psychological and Brain Sciences Boston University <i>samling@bu.edu</i>